THE SCHEME THEORETIC IMAGE IS FUNCTORIAL ALONG A COMMUTATIVE DIAGRAM OF MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-factor-factor is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 948-967 and proof span 969-974 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains two xymatrix diagrams; both are copied verbatim from the counted statement span, and the retained chapter preamble loads \usepackage[all]{xy}, so no diagram package was added on our side.

Editorial note. The proof is the author's complete argument and ends with the author's own closing sentence 'This proves the lemma.', which is kept verbatim.

Allowed standard prerequisites (editorial).

- Scheme theoretic image: for a morphism f_i : X_i to Y_i the image Z_i is the smallest closed subscheme of Y_i through which f_i factors. This is the defining property used at share of proof lines 969-972. Supplies the universal property which is applied to the pullback of Z_2 in Y_1.
- Scheme theoretic inverse image of a closed subscheme along a morphism, and the fact that a morphism factors through it exactly when it factors through the original closed subscheme. Used at proof lines 969-971 to produce a closed subscheme of Y_1 through which f_1 factors. Produces the closed subscheme of Y_1 that contains Z_1 by minimality, which is the whole argument.
- Xy-pic commutative diagrams. The statement and its conclusion are typeset as xymatrix displays at statement lines 949-956 and 958-966. The two displayed squares are the author's own xymatrix diagrams; the shared Stacks preamble retained for this chapter loads Xy-pic, so nothing is retypeset.

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Lemma 0.1. *Let*

$$\begin{array}{ccc} X_1 & \xrightarrow{f_1} & Y_1 \\ \downarrow & & \downarrow \\ X_2 & \xrightarrow{f_2} & Y_2 \end{array}$$

be a commutative diagram of schemes. Let $Z_i \subset Y_i$, $i = 1, 2$ be the scheme theoretic image of f_i. Then the morphism $Y_1 \to Y_2$ induces a morphism $Z_1 \to Z_2$ and a commutative diagram

$$\begin{array}{ccccc} X_1 & \longrightarrow & Z_1 & \longrightarrow & Y_1 \\ \downarrow & & \downarrow & & \downarrow \\ X_2 & \longrightarrow & Z_2 & \longrightarrow & Y_2 \end{array}$$

Proof. The scheme theoretic inverse image of Z_2 in Y_1 is a closed subscheme of Y_1 through which f_1 factors. Hence Z_1 is contained in this. This proves the lemma. □